

TADP-Sec

Trustworthy AI Data Preparation with Governance-to-Runtime Security for Federated Learning

Supplementary Material

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Purpose of This Supplement

This supplementary document provides methodological, experimental, and reproducibility details supporting the accompanying TADP-Sec paper.

The main manuscript focuses on the framework, principal experimental design, and central findings. This supplement provides additional details on the TADP governance evidence rubric and admission policy; the TADP-Sec security-profile and session-selection policy; the complete federated-learning and protected-aggregation configuration; governance sensitivity and ablation analyses; extended utility, operational, protocol-enforcement, and availability results; and metric-equivalence, non-inferiority, and reproducibility diagnostics.

The supplementary analyses do not modify the frozen policy or the reported canonical experiment. Governance sensitivity and ablation analyses are post-freeze robustness analyses and were not used to retune the reported admission thresholds or weights.

S1. Governance Evidence, Policy, and Controlled Archetypes

S1.1 Governance Dimensions and Scoring

TADP evaluates each governed contribution across six governance dimensions:

- Source Reliability (SR)
- Data Quality (DQ)
- Documentation (DOC)
- Timeliness (TIME)
- Regulatory Alignment (REG)
- Context of Use (CTX)

The financial-fraud experimental profile uses the fixed policy weights shown below.

Governance dimension	Weight (%)	Critical dimension
Source Reliability (SR)	0.20	Yes
Data Quality (DQ)	0.25	Yes
Documentation (DOC)	0.15	No
Timeliness (TIME)	0.15	No
Regulatory Alignment (REG)	0.15	Yes
Context of Use (CTX)	0.10	Yes

These weights are frozen policy parameters for the reported financial-fraud experiment. They are not claimed to be universal or empirically optimal across application domains. Alternative deployments may define and version their own weights, thresholds, and critical-dimension sets while retaining the same TADP admission mechanism.

S1.2 Common Evidence Rubric

All governance dimensions use a common 0–5 semantic scale.

Score	Policy level
0	No Evidence
1	Very Weak
2	Limited
3	Adequate
4	Strong
5	Very Strong

Data owners or stewards provide structured, evidence-supported conditions rather than authoritative numerical trust scores. The server validates the submitted evidence and maps the validated conditions to the frozen rubric.

Source Reliability: source assurance, data-controller identity, and collection objective.

Documentation: data dictionary, version history, collection protocol, and definition updates.

Timeliness: freshness requirements, refresh schedules, and retention requirements.

Regulatory Alignment: lawful basis, consent or ethical approval, geographic restrictions, sensitivity classification, and audit status.

Context of Use: intended-use compatibility, population or domain coverage, known limitations, licensing permissions, and redistribution constraints.

Data Quality is treated differently. It is measured technically from each client's actual local dataset partition by the Local Evidence Validator rather than being supplied by the controlled documentary archetype.

S1.3 Overall HPS

Let $D = \{SR, DQ, DOC, TIME, REG, CTX\}$ denote the complete set of governance dimensions. For client i , the overall governance score is:

$$HPS_i = \sum_{d \in D} w_d D_{i,d}$$

where w_d is the frozen weight associated with governance dimension d and $D_{i,d}$ is the validated score of client i for dimension d . Because the weights sum to 1, HPS remains on the common 0–5 semantic scale.

S1.4 Critical-Dimension Safeguards

The critical governance set is $C = \{SR, DQ, REG, CTX\}$. The total weight assigned to the critical dimensions is 0.70.

$$HPS_{C,i} = [\sum_{d \in C} w_d D_{i,d}] / [\sum_{d \in C} w_d]$$

Normalizing by the critical-weight total keeps HPS_C on the same 0–5 scale as the overall HPS.

$$C_{i,min} = \min_{d \in C} D_{i,d}$$

The frozen policy parameters are:

- $T_R = 3.0$ — lower HPS admission boundary
- $T_A = 4.0$ — direct-acceptance HPS boundary

- $T_{C,direct} = 4.0$ — direct-acceptance critical-profile boundary
- $T_{C,review} = 3.5$ — automated-review critical-profile boundary
- $C_{min} = 3.0$ — minimum acceptable score for every critical dimension

$$T_{C,review} = (T_R + T_A) / 2 = 3.5$$

This value is policy-derived and is not selected to obtain a predetermined admission percentage.

S1.5 Frozen Admission Rule

1. Automatic rejection. If $HPS_i < T_R$, the contribution is rejected.

2. Non-compensatory critical-dimension safeguard. If $C_{i,min} < C_{min}$, the contribution is routed to remediation. A high overall HPS or HPS_C cannot compensate for an individual critical dimension below the minimum acceptable level.

3. Direct acceptance. If $HPS_i \geq T_A$ and $HPS_{C,i} \geq T_{C,direct}$ and every critical dimension satisfies C_{min} , the contribution is directly accepted.

4. Deterministic automated-review acceptance. If $T_R \leq HPS_i < T_A$, the contribution enters the deterministic automated-review path. It is accepted only if $HPS_{C,i} \geq T_{C,review}$ and every critical dimension satisfies C_{min} .

5. Remaining cases. Remaining contributions are routed to remediation.

The automated-review path uses the same already validated governance scores. It is not a discretionary reassignment or subjective rescoring of HPS.

S1.6 Evidence Integrity and Structural Prerequisites

Governance concepts already represented inside the HPS dimensions are not duplicated as a second scored mandatory-gate layer. For example, lawful basis, regulatory evidence, and intended-use compatibility are validated and scored within their corresponding governance dimensions.

Structural evidence-integrity requirements are treated separately. These include valid cryptographic authentication, protocol/version compatibility, and binding between submitted evidence and the governed dataset commitment. Failure of these structural requirements prevents the affected evidence package from being treated as valid governance evidence. Such failures cannot be compensated for by a high HPS or HPS_C .

S1.7 Controlled Governance Archetypes

The public ULB credit-card-fraud dataset contains transaction-level features and class labels but does not contain organization-level governance evidence. Accordingly, five controlled archetypes are used for the nontechnical governance dimensions. Data Quality remains independently measured from each actual local client partition.

Profile	SR	DOC	TIME	REG	CTX
G1 Mature	5	4	4	5	4
G2 Strong	4	4	4	4	4
G3 Adequate	3	3	3	3	3
G4 Remediation	2	3	3	2	2
G5 Weak	1	2	2	1	1

The fixed 20-client experimental fixture contains 3 G1 clients, 3 G2 clients, 6 G3 clients, 4 G4 clients, and 4 G5 clients. Client identifiers are assigned non-monotonically so that identifier order does not encode governance maturity.

These proportions define the controlled experimental fixture only. They are not admission quotas and are not used to derive the TADP thresholds. Final admission outcomes are determined solely by the frozen admission policy applied to the validated evidence, including Data Quality measured from each client's actual partition.

S2. TADP-Sec Security Profile and Session Specification

S2.1 Separation Between Governance Admission and Runtime Protection

TADP and TADP-Sec answer two different questions. TADP asks whether a contribution is sufficiently governed to participate. TADP-Sec asks what minimum runtime protection is required for an admitted contribution. CIA and Business Impact are therefore evaluated only after TADP admission and remain separate from the HPS calculation.

S2.2 CIA High-Water Classification

For admitted client i , let C_i , I_i , and A_i denote confidentiality, integrity, and availability impact, respectively. The CIA high-water value is:

$$h_i = \max(C_i, I_i, A_i)$$

The four impact levels are Low, Moderate, High, and Critical. The Low, Moderate, and High terminology follows impact-based information-security categorization principles. The Critical level is a TADP-Sec policy extension used by the framework. The exact CIA $\times$ Business-Impact matrix below is a TADP-Sec policy design and should not be interpreted as a security-tier matrix defined by NIST.

S2.3 CIA and Business-Impact Mapping

CIA high-water \ Business Impact	Low	Moderate	High	Critical
Low	C1	C2	C3	C4
Moderate	C2	C2	C3	C4
High	C3	C3	C4	C4
Critical	C4	C4	C4	C4

$$C1 \rightarrow T1 \quad C2 \rightarrow T2 \quad C3 \rightarrow T3 \quad C4 \rightarrow T4$$

The mapping is deterministic for a fixed policy version.

S2.4 Session-Wide Protection Policy

Let A denote the closed admitted cohort. Each admitted client receives a minimum required runtime profile T_i . One common session profile is selected according to:

$$T_{\text{session}} = \max_{i \in A} T_i$$

The strictest admitted-client requirement determines the protection profile for the complete session. The decision is made before federated round 1. Once the session begins, the admitted cohort is closed, the selected profile remains fixed, the profile is not downgraded, and the session does not dynamically switch between $T1$ – $T4$.

A new client, a materially changed dataset commitment, or a materially changed governance/security policy requires reassessment and formation of a new session.

In the canonical experiment, the 12 admitted clients required $T2$ for 3 clients, $T3$ for 4 clients, and $T4$ for 5 clients. The strictest admitted requirement was therefore $T4$. $T1$ – $T3$ are retained as benchmark profiles over the same admitted cohort to compare the implemented protection mechanisms. They are not alternative profiles dynamically selected within the reported $T4$ -governed session.

S2.5 Runtime Protection Profiles

Tier	Protection profile
T1	Authenticated governed federated learning with signature verification, freshness and anti-replay checks, challenge-bound participation control, session binding, and tamper-evident audit logging. T1 does not use encrypted aggregation.
T2	Modified Domingo-Ferrer protected transformation with matrix key switching.
T3	xMK-CKKS multi-key encrypted aggregation with all-participant collaborative decryption.
T4	SAMK multi-key BFV aggregation with Paillier-protected helper information and independent authorized-client recovery.

The tier number represents a policy protection profile, not a claim of monotonically increasing standardized cryptographic bit security. $T2$, $T3$, and $T4$ use distinct protected-aggregation mechanisms and are not intended to represent one nested cryptographic stack.

S2.6 Protocol Participation Control

TADP-Sec includes a challenge-bound Proof-of-Flow/Challenge (PoFC) participation control. PoFC binds an authenticated client submission to server-issued session and round context and enforces the permitted

submission limit for that challenge. PoFC is a protocol-level participation control. It is not presented as a general network-level denial-of-service defense.

S3. Complete Experimental and Cryptographic Configuration

S3.1 Dataset

The canonical experiment uses the full ULB credit-card-fraud dataset.

- Total transactions: 284,807
- Fraud transactions: 492
- Normal transactions: 284,315
- Fraud prevalence: approximately 0.173%

The data are divided using a two-stage stratified split:

- Global held-out test set: 56,962 records
- Validation set: 45,569 records
- Federated client-training pool: 182,276 records

The client-training pool contains 315 fraud records. The validation and test sets remain outside client training.

S3.2 Federated Client Partitioning

The training pool is distributed across 20 non-IID federated clients using class-conditional Dirichlet partitioning with Fraud $\alpha = 0.80$ and Normal $\alpha = 1.20$. The canonical partition is controlled by the fixed client-partition seed used by the experiment. No validation or test records participate in the client partition.

S3.3 Feature Preprocessing

A common StandardScaler is fitted exclusively on the union of the client-training records in the simulation. The fitted scaler is then applied to every client's training partition, the validation set, and the held-out test set. Validation and test records are not used to estimate scaler parameters. This design maintains a common model coordinate system across clients in the experimental simulation. It is not proposed as a production federated normalization protocol; a real deployment would require an appropriate privacy-preserving or federated statistic-estimation mechanism.

S3.4 Federated-Learning Configuration

Parameter	Value
Dataset	ULB credit-card fraud
Transactions	284,807
Fraud transactions	492
Global test set	56,962
Validation set	45,569
Client-training pool	182,276
FL clients	20
Fraud Dirichlet α	0.80
Normal Dirichlet α	1.20
Seeds	7, 11, 19, 23, 42
Federated rounds	20
Local epochs	2
Batch size	32
Learning rate	0.01
L2 regularization λ	0.005
FPR constraint	0.005
AP non-inferiority margin	$\delta = 0.005$
Aggregation	Sample-size-weighted FedAvg

A regularized logistic-regression model is used for the financial-fraud task. For valid participant i , FedAvg uses:

$$\alpha_i = n_i / \sum_j n_j$$

$$\Delta = \sum_i \alpha_i \Delta_i$$

where the denominator includes the valid current-round participants. Clients removed before aggregation are excluded from this denominator.

S3.5 Classification Threshold Selection

The model-classification threshold is selected on the validation set subject to a maximum false-positive-rate constraint $FPR \leq 0.005$. The selected threshold is then applied to the held-out test data. The held-out test set is therefore used only for final evaluation and does not participate in threshold selection.

S3.6 T2 Configuration

T2 uses a modified Domingo-Ferrer protected-transformation reference implementation with matrix key switching. The principal experimental parameters are:

- Transformation dimension: $d = 50$
- Fixed-point scale: 10^6
- Base modulus: $m_0 = 2^{61} - 1$
- Extended modulus: $m = m_0(2^{67} - 1)$
- Matrix key-switching noise bound: 3

T2 uses a deterministic, data-independent numerical preflight before the federated experiment begins. The preflight uses update dimension 30; scales 10^{-4} , 10^{-3} , and 10^{-2} ; and synthetic client weights 0.10, 0.20, 0.30, and 0.40.

S3.7 T3 Configuration

- Polynomial degree: $N = 4096$
- $\log_2(q)$: 180

- CKKS scale: 2^{30}
- CKKS noise standard deviation: 3.2
- Collaborative decryption-share noise standard deviation: 8.0
- Decryption policy: all valid participating clients

The implementation uses all-participant collaborative decryption and is not presented as a configurable t-of-n threshold-decryption implementation.

S3.8 T4 Configuration

T4 follows the SAMK workflow at the algorithmic level using multi-key BFV, Paillier-protected helper information, and polynomial interpolation. The reference parameters are:

- BFV polynomial degree: $N = 64$
- Ciphertext modulus q : 35,184,372,088,777
- Plaintext modulus t : 16,777,213
- Fixed-point scale: 10^6
- BFV noise bound: 1
- Paillier prime size: 320 bits per prime
- Resulting Paillier modulus: approximately 640 bits

These parameters are selected to make the exact-integer Python reference implementation computationally tractable. They are functional and reproducibility parameters, not a production bit-security configuration and not an optimized SEAL/OpenFHE performance configuration.

S3.9 T4 Protected-Aggregation Workflow

Each participating client independently owns a BFV key pair and a Paillier key pair. Before encryption, the client's local model update is multiplied by its sample-size FedAvg weight:

$$\mathbf{u}_i = \alpha_i \Delta_i$$

The complete update includes both model-weight deltas and the bias delta. Each client encrypts its weighted update under its own BFV key.

For uploader i , the client constructs a key-dependent helper polynomial:

$$f_i(\mathbf{X}) = \mathbf{s}_i \cdot \mathbf{c}_{i,1}(\mathbf{X})$$

where $\mathbf{s}_i$ is the client's BFV secret key and $\mathbf{c}_{i,1}$ is the second BFV ciphertext component. The helper polynomial is evaluated at public interpolation points. For the first $N-1$ helper points, each uploader encrypts its helper value using the server's Paillier public key. For the final helper point, each uploader encrypts its helper value separately for every authorized recipient using that recipient's Paillier public key.

The server aggregates the BFV c_0 components; homomorphically adds the first $N-1$ Paillier-protected helper values and decrypts only their aggregate values; creates a recipient-specific encrypted aggregate of the final helper value; and packages the aggregate information for authorized participants. The server therefore has only $N-1$ plaintext aggregate helper points and does not receive the complete helper information required for independent aggregate reconstruction.

Each authorized participating recipient decrypts its recipient-specific final aggregate helper point, combines it with the $N-1$ aggregate helper points, interpolates the aggregate helper polynomial, and reconstructs the

combined weighted FedAvg update. The recovered result is the aggregate $\sum_i \alpha_i \Delta_i$ rather than an individual client's update.

S3.10 Numerical Fidelity Gates

Protected aggregation is evaluated against an experiment-side plaintext FedAvg reference. The plaintext reference is used only for fidelity validation and is not exposed to the protected aggregation-server interface. A protected round passes only when recovered and reference vectors have compatible shapes, all required values are finite, maximum absolute coordinate error remains below the scenario-specific limit, and RMSE remains below the scenario-specific limit.

Scenario	Maximum absolute error	RMSE
B1 xMK-CKKS	2×10^{-4}	5×10^{-5}
T2 DF+KS	2×10^{-6}	5×10^{-7}
T3 xMK-CKKS	2×10^{-4}	5×10^{-5}
T4 SAMK	1×10^{-4}	1×10^{-4}

Relative L2 error is recorded as a diagnostic but is not used as the hard acceptance criterion because the reference update norm can become very small near convergence. Failure of a required fidelity gate aborts the protected round. A failed protected aggregate is not replaced by the plaintext reference.

S4. Governance Sensitivity and Ablation

S4.1 Purpose and Scope

The governance robustness analyses evaluate the sensitivity of the frozen TADP admission policy. They are post-freeze stress tests rather than a search for alternative policy parameters. The analyses do not rerun federated training, do not modify the canonical results, do not retune the reported governance weights, and do not retune the admission thresholds. Before applying perturbations, the analysis independently replayed the stored governance records and reproduced the canonical baseline of 12 admitted clients.

S4.2 Threshold Sensitivity

The admission thresholds are perturbed by -30% , -20% , -10% , 0% , $+10\%$, $+20\%$, and $+30\%$. Three conditions are evaluated: change T_R only; change T_A only; and change T_R and T_A jointly. The critical-dimension floor remains fixed at $C_{\min} = 3$ because it represents the categorical Adequate minimum rather than a continuously tuned parameter.

$$T_{C,direct} = \text{perturbed } T_A$$

$$T_{C,review} = (T_R + T_A) / 2$$

Perturbation	-30%	-20%	-10%	0%	+10%	+20%	+30%
Lower threshold only	12	12	12	12	11	6	6
Upper threshold only	Invalid	12	12	12	11	6	6
Joint thresholds	12	12	12	12	6	6	3

The upper-only -30% condition is invalid because it would produce $T_A < T_R$. Relaxing the thresholds did not add new clients to the admitted cohort. Clients excluded under the baseline policy continued to be constrained by the critical-profile and critical-dimension requirements. In contrast, tightening the policy reduced participation. Increasing both T_R and T_A by 10% reduced the admitted cohort from 12 to 6

clients, while the +30% joint stress condition retained only 3 clients. These step changes reflect the clustered governance profiles in the controlled 20-client fixture and should not be generalized to other client populations.

S4.3 Weight Sensitivity

Each governance dimension is perturbed separately by −30%, −20%, −10%, 0%, +10%, +20%, and +30%. After changing the target weight, the remaining five weights are proportionally rescaled so that $\sum_{d \in D} w_d = 1$. The complete admission policy is recomputed, including overall HPS, normalized critical HPS_C, and the individual critical-dimension floor.

Across the tested one-at-a-time perturbations, only one condition changed the binary admitted set: reducing the Data Quality weight by 30%. The DQ weight decreased from 0.25 to 0.175 and the admitted cohort changed from 12 to 11 clients. No other tested one-at-a-time weight perturbation changed the binary admission outcome. Within the tested range, the admission decision was therefore more stable to isolated weight perturbations than to tightening of the policy thresholds.

S4.4 Leave-One-Dimension-Out Ablation

Leave-one-dimension-out (LODO) ablation evaluates the effect of removing an entire governance dimension. For an omitted dimension d , the remaining weights are renormalized according to:

$$w_j^{(-d)} = w_j / (1 - w_d), \text{ for } j \neq d$$

If the omitted dimension belongs to the critical set, it is also removed from the critical set before recalculating HPS_C and C_min. The complete admission policy is then replayed.

Omitted dimension	Admitted	Admission flips	Jaccard similarity	Mean ΔHPS
Data Quality	6	6	0.500	0.5050
Source Reliability	12	0	1.000	0.1762
Regulatory Alignment	12	0	1.000	0.1244
Context of Use	12	0	1.000	0.0817
Documentation	12	0	1.000	0.0593
Timeliness	12	0	1.000	0.0593

Removing Data Quality produced the largest observed admission-level effect: the admitted cohort decreased from 12 to 6 clients, admission flips were 6, and admitted-set Jaccard similarity fell to 0.50. Removing any other single dimension altered HPS values but did not change the binary admitted set.

This finding should not be interpreted as a universal ranking of governance dimensions. The stronger DQ effect is partly structural to this experiment because DQ has the largest baseline policy weight (0.25), belongs to the critical set, and is measured directly from heterogeneous client partitions, while the other nontechnical dimensions use controlled governance archetypes.

S4.5 Joint Threshold–Weight Sensitivity

The joint analysis tests whether threshold and weight changes interact. For each governance dimension, seven joint threshold perturbations are crossed with seven one-at-a-time weight perturbations, producing a 7×7 sensitivity surface.

$$\Delta N_{\text{admitted}} = N_{\text{admitted}}^{(\delta T, \delta w)} - N_{\text{admitted}}^{(0,0)}$$

$$N_{\text{admitted}}^{(0,0)} = 12$$

For Source Reliability, Documentation, Timeliness, Regulatory Alignment, and Context of Use, the response surfaces were dominated by the threshold perturbation. Within the tested range, changing these weights did not add an admission effect beyond the threshold condition. Data Quality was the only dimension with an additional weight-dependent interaction. At the baseline thresholds, reducing the DQ weight by 30% reduced admission by one client. Under the +10% joint-threshold condition, increasing the DQ weight by 30% reduced the admission loss from six clients to one. These observations describe the structure of the controlled cohort and were not used to retune the frozen governance policy.

S5. Extended Performance and Security Results

S5.1 Experimental Scenarios

- B0: Plain federated learning
- B1: Plain-cohort xMK-CKKS baseline
- A1: TADP admission-only ablation
- T1: Governed authenticated FL
- T2: TADP + modified DF/KS
- T3: TADP + xMK-CKKS
- T4: TADP + SAMK

B0 and B1 use all 20 clients. A1 and T1–T4 use the same fixed 12-client TADP-admitted cohort. All main scenarios use the same five seeds and 20 federated rounds.

S5.2 Main Operational Comparison

Scenario	AP	F1	MCC	Runtime/seed (s)	Crypto ms/round	Generated payload (KB/round)	Max abs. error	Max RMSE
B0 Plain FL	0.7280	0.7197	0.7311	18.3	0	9.38	—	—
B1 xMK-CKKS	0.7280	0.7197	0.7311	570.3	27,458.38	9,600.00	1.11×10^{-4}	2.98×10^{-5}
A1 TADP	0.7267	0.7186	0.7276	10.2	0	5.62	—	—
T1 Plain TADP	0.7267	0.7186	0.7276	23.4	0	5.62	—	—
T2 DF+KS	0.7267	0.7186	0.7276	161.9	6,891.10	284.06	9.64×10^{-7}	2.06×10^{-7}
T3 xMK-CKKS	0.7267	0.7186	0.7276	353.5	16,444.56	5,760.00	6.40×10^{-5}	1.91×10^{-5}
T4 SAMK	0.7269	0.7186	0.7276	116.2	4,660.50	186.98	3.75×10^{-5}	2.42×10^{-5}

Runtime is the mean end-to-end execution time per seed. Generated protocol payload represents the byte volume of the generated protocol objects; it is not a wire-level network capture. Every protected clean scenario completed all 100/100 required round-level numerical fidelity checks without failure. T4 additionally completed 1,200/1,200 authorized-recipient aggregate recoveries successfully.

S5.3 Runtime and Payload Relative to T1

- T2 runtime: approximately $6.92 \times T1$
- T3 runtime: approximately $15.11 \times T1$
- T4 runtime: approximately $4.97 \times T1$
- T2 generated payload: approximately $50.54 \times T1$

- T3 generated payload: approximately $1,024.9 \times T1$
- T4 generated payload: approximately $33.27 \times T1$

These measurements characterize the current reference implementations and should not be interpreted as a ranking of standardized cryptographic strength. In particular, T4's lower measured runtime relative to T2 and T3 is implementation-specific. The profiles use structurally different mechanisms.

S5.4 Protocol-Enforcement Tests

Profile	Injected violation	Injected	Blocked	Benign blocked	Effective clients
T1	Invalid signature	2	2/2	0	10
T2	Replay/stale update	1	1/1	0	11
T3	PoFC submission overflow	3	3/3	0	9

All injected violations were rejected before aggregation. These tests demonstrate enforcement of the implemented signature/authentication controls, freshness and anti-replay controls, and challenge-bound participation controls. They do not establish general robustness against semantic data poisoning, backdoor attacks, arbitrary Byzantine behavior, or general network-level denial-of-service attacks.

S5.5 T4 Dropout and Delayed-Recovery Experiment

A separate T4 availability experiment evaluates pre-upload dropout and post-upload delayed recipient recovery. The experiment uses 5 seeds, 5 rounds per seed, 2 pre-upload dropouts, and 10 surviving uploaders. Clients unavailable before upload are excluded from the protected survivor aggregate, and the FedAvg denominator is recomputed over the current valid participants. The protocol session is re-formed for the surviving client set while preserving the locked T4 profile. A client that has already completed upload may temporarily disconnect without preventing the remaining authorized recipients from recovering the aggregate. When the delayed uploader reconnects, it may recover the same already-formed aggregate package.

Measure	Result
Pre-upload dropouts	2
Final effective clients	10
Delayed-recipient tests	25
Successful delayed recovery	25/25
All authorized recipients passed	Yes
Protected fidelity passed	Yes
Session re-formed	Yes
Tier downgraded	No
Server recovered plaintext aggregate	No

The dropout experiment evaluates an availability property of the T4 reference workflow. It is not presented as a production service-level availability guarantee.

S6. Metric Equivalence, Non-Inferiority, and Reproducibility

S6.1 Purpose of Full-Precision Diagnostics

The main manuscript reports AP, F1, and MCC at conventional reporting precision. Because protected T2–T4 aggregation recovers approximations to the corresponding plaintext FedAvg computation, rounded utility values alone can hide small non-zero parameter or prediction-score differences. The full-precision diagnostics therefore distinguish practical utility equivalence from numerical identity.

For each seed, the final models are compared using the planned references: B1 vs B0, T1 vs A1, T2 vs T1, T3 vs T1, and T4 vs T1. The diagnostics include Δ AP, Δ F1, Δ MCC, final-model parameter L2 distance, maximum parameter difference, maximum test-probability difference, probability RMSE, and binary prediction disagreements.

S6.2 Full-Precision Metric-Equivalence Results

Comparison	Mean Δ AP	Max $ \Delta$ F1	Max $ \Delta$ MCC	Mean model L2 distance	Max $ \Delta$ probability	Prediction disagreements
B1 vs B0	1.10×10^{-7}	0	0	2.06×10^{-4}	2.87×10^{-4}	0
T1 vs A1	0	0	0	0	0	0
T2 vs T1	4.93×10^{-9}	0	0	1.72×10^{-6}	2.03×10^{-6}	0
T3 vs T1	3.01×10^{-7}	0	0	1.45×10^{-4}	2.05×10^{-4}	0
T4 vs T1	1.48×10^{-4}	0	0	1.33×10^{-3}	1.53×10^{-3}	0

The protected profiles are therefore not numerically identical to their references. However, the numerical perturbations did not change the reported F1 or MCC and produced no binary prediction disagreements across the evaluated final test predictions. Small parameter or probability differences can leave AP effectively unchanged when the ranking of test samples is preserved. Likewise, threshold-based F1 and MCC remain unchanged when no prediction crosses the selected classification threshold.

S6.3 Planned AP Non-Inferiority Analysis

Average Precision is compared seed-by-seed using the same five seeds: 7, 11, 19, 23, and 42. For candidate scenario c and reference scenario r:

$$\Delta AP_s = AP_{c,s} - AP_{r,s}$$

The frozen non-inferiority margin is $\delta = 0.005$. The candidate is considered non-inferior when the one-sided 95% lower confidence bound for ΔAP is greater than -0.005 . Passing non-inferiority does not require the candidate to outperform the reference; it means that the data support the conclusion that the candidate's AP degradation does not exceed the predefined acceptable loss boundary. The margin was frozen before the canonical full-dataset run.

S6.4 Planned AP Contrasts

Comparison	Mean ΔAP	One-sided 95% lower bound	Boundary	Non-inferior
B1 vs B0	1.10×10^{-7}	-2.40×10^{-7}	-0.005	Yes
A1 vs B0	-1.27×10^{-3}	-1.72×10^{-3}	-0.005	Yes
T1 vs A1	0	0	-0.005	Yes
T2 vs T1	4.93×10^{-9}	-5.14×10^{-9}	-0.005	Yes
T3 vs T1	3.01×10^{-7}	-1.05×10^{-7}	-0.005	Yes
T4 vs T1	1.48×10^{-4}	-1.87×10^{-4}	-0.005	Yes

All six planned contrasts satisfy the predefined AP non-inferiority criterion. For the governance effect specifically, A1 vs B0 has mean $\Delta AP \approx -0.00127$ and one-sided lower bound ≈ -0.00172 . Since $-0.00172 > -0.005$, the governed 12-client cohort is considered non-inferior to the 20-client plain-FL reference under the predefined AP margin. This finding should be interpreted as evidence that the observed utility reduction remains within the predefined acceptable loss, not as evidence that governance improves AP.

S6.5 Audit and Reproducibility Scope

The experimental implementation records governance and runtime artifacts needed to reconstruct the reported decisions, including experiment and protocol versions; dataset identity and manifest information; dataset commitments; client-partition configuration; governance-policy identifiers; evidence-validation outcomes; HPS and critical-profile values; admission decisions; CIA/Business-Impact classifications; session-tier decisions; aggregate-fidelity diagnostics; protocol-enforcement outcomes; T4 availability results; AP non-inferiority outputs; metric-equivalence diagnostics; and tamper-evident ledger records and final seals.

The audit layer does not store raw client datasets, private cryptographic keys, or plaintext individual client model updates. The canonical full experimental run produced 22 final ledger seals.

S6.6 Canonical Dataset Identifier

The canonical ULB credit-card-fraud dataset used for the reported experiment has SHA-256:

76274b691b16a6c49d3f159c883398e03ccd6d1ee12d9d8ee38f4b4b98551a89

It contains 284,807 transactions, 492 fraud records, and 284,315 non-fraud records. A dataset that does not match the canonical experiment should not be described as an exact reproduction of the reported full-dataset results.

S6.7 Governance Policy Identifier

financial_fraud_experimental_profile_v1

The controlled governance archetypes are used because the public transaction dataset does not contain organizational governance metadata. Data Quality remains technically derived from the actual client partitions.

S6.8 Source-Code Reproducibility

The exact source used for the canonical experimental execution should be archived with the repository as the canonical executed source.

TADP_Sec_v11_7_FINAL_v3_2_fresh_run_ledger_reset_metric_reporting_fixed.py

SHA-256:

4aeb27a65c36a21a57c63182ed418e575fdac34dea8ecebc1113f8e0b80d9850

The governance sensitivity/ablation implementation was:

TADP_v11_7_FINAL_governance_sensitivity_ablation_v3_2_no_autodownload.py

SHA-256:

247422bb453ac5d17cd5952100117242bb4c3e29fdddc567e7fd789e06e496fd

S7. Scope and Interpretation Notes

Governance evidence. The public financial-fraud dataset does not contain real institutional governance documentation. Therefore, nontechnical governance and CIA/Business-Impact conditions are represented by reproducible controlled archetypes. The experiment demonstrates the operation of the governance-to-runtime workflow but does not establish that the particular governance weights or thresholds are universally optimal.

Cryptographic implementations. T2–T4 are research/reference implementations. The results demonstrate implementation functionality, protected-aggregate numerical fidelity, experimental runtime, generated protocol-object payload, protocol-enforcement behavior, and the evaluated T4 recovery workflow. They do not constitute cryptographic certification of a production deployment.

T4 parameterization. The T4 BFV and Paillier parameters are functional/reproducibility settings selected for tractable pure-Python execution. They are not presented as a production bit-security configuration.

Protocol-enforcement scope. The injected invalid-signature, replay/staleness, and PoFC violations evaluate specific protocol controls. The study does not claim comprehensive defense against poisoning, backdoor, Byzantine, or network-level denial-of-service attacks.

Runtime measurements. Runtime values are measured from the reference experimental implementation and should not be generalized directly to optimized production cryptographic libraries or distributed hardware deployments.

Communication measurements. Reported communication values correspond to generated protocol-object byte volume. They are not packet-level network traces and do not include all transport-layer overhead.

Energy and carbon. The canonical experiment was executed in a dynamic computational environment that does not provide a stable basis for reproducible absolute hardware-energy or carbon measurement. Therefore, the study reports runtime and generated protocol payload rather than inferring absolute energy or carbon values from generic hardware assumptions.

S8. Repository Artifacts

The GitHub repository accompanying the study should contain, at minimum:

- Canonical experiment: TADP-Sec_v11.7_Canonical_Experiment.ipynb
- Exact executed source corresponding to the reported canonical run
- Supplement: TADP-Sec_Supplementary_Material.pdf

- Main experimental results: admission results, B0/B1/A1/T1–T4 operational results, and aggregate-fidelity diagnostics
- Protocol and availability results: protocol-enforcement results and T4 dropout/delayed-recovery results
- Statistical diagnostics: AP non-inferiority planned contrasts, metric-equivalence summary, and per-seed metric-equivalence outputs
- Governance robustness: threshold sensitivity, weight sensitivity, leave-one-dimension-out ablation, and joint threshold–weight results/heatmaps
- Reproducibility: dataset manifest, relevant version/configuration records, and audit summaries/final ledger seals

The original ULB credit-card-fraud CSV should not be redistributed in the repository. Users should obtain the dataset from its authorized original/public source.

End of Supplementary Material